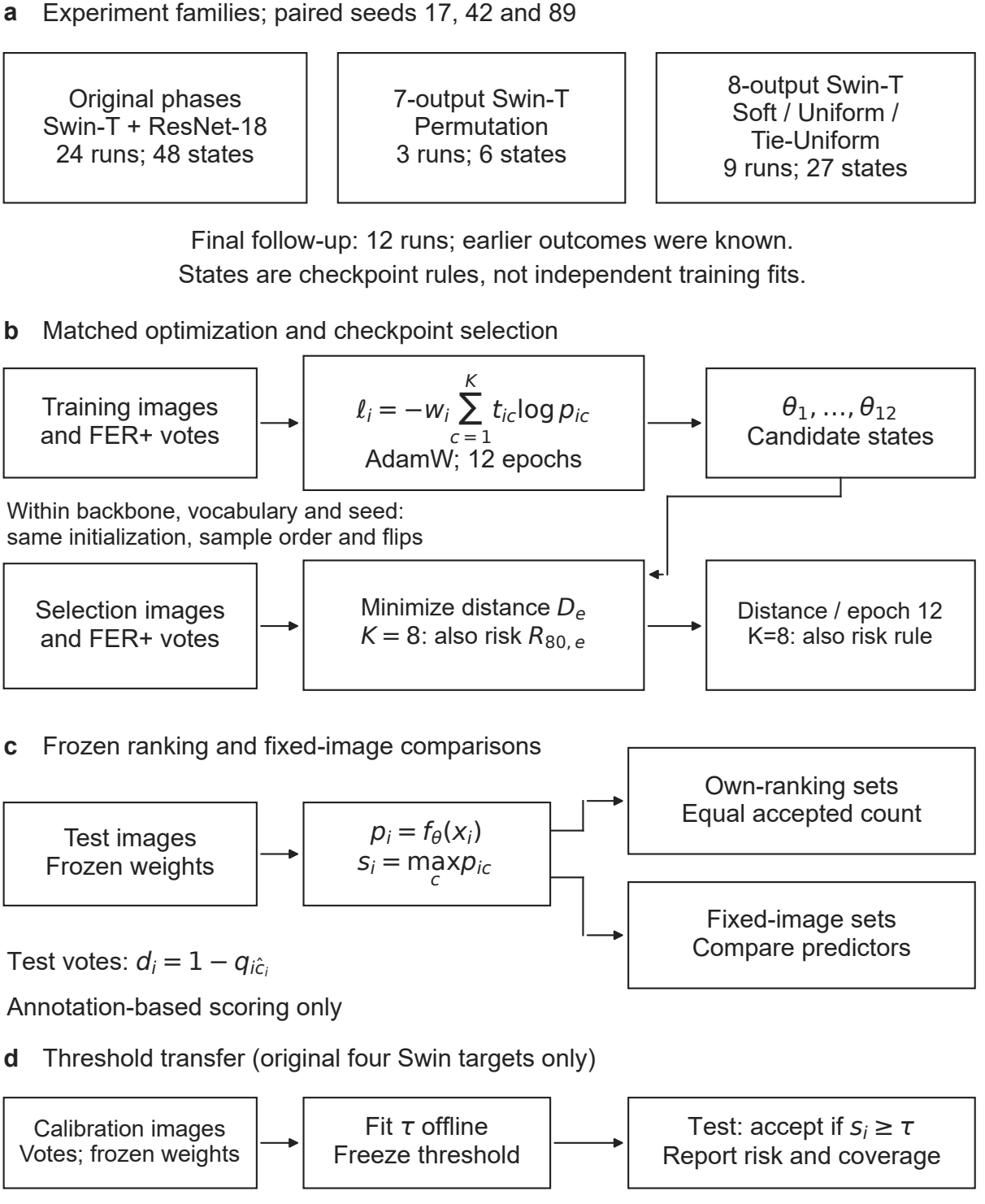

a Experiment families; paired seeds 17, 42 and 89
Original phases
Swin-T + ResNet-18
24 runs; 48 states
7-output Swin-T
Permutation
3 runs; 6 states
8-output Swin-T
Soft / Uniform /
Tie-Uniform
9 runs; 27 states
Final follow-up: 12 runs; earlier outcomes were known.
States are checkpoint rules, not independent training fits.
b Matched optimization and checkpoint selection
Training images
and FER+ votes
$\ell_i = -w_i \sum_{c=1}^{K} t_{ic} \log p_{ic}$
AdamW; 12 epochs
$\theta_1, \ldots, \theta_{12}$
Candidate states
Within backbone, vocabulary and seed:
same initialization, sample order and flips
Selection images
and FER+ votes
Minimize distance D_e
$K = 8$: also risk $R_{80,e}$
Distance / epoch 12
K=8: also risk rule
c Frozen ranking and fixed-image comparisons
Test images
Frozen weights
$p_i = f_\theta(x_i)$
$s_i = \max_c p_{ic}$
Own-ranking sets
Equal accepted count
Fixed-image sets
Compare predictors
Test votes: $d_i = 1 - q_{i\hat{c}_i}$
Annotation-based scoring only
d Threshold transfer (original four Swin targets only)
Calibration images
Votes; frozen weights
Fit τ offline
Freeze threshold
Test: accept if $s_i \geq \tau$
Report risk and coverage